

BrightLeam

Exercise 2 SQL Aggregated ~~Database~~

Students table

Student_id	name	age	department
1	Alice	20	IT
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	finance
5	Eve	22	HR

Q1 Distinct department in the Students table

Query: `SELECT DISTINCT department
from Students`

department
IT
finance
HR

Question 2

Get the average age of Students per department

Query: `SELECT department,
AVG(age) AS avg_age
from Students
GROUP BY departments`

department	avg_age
IT	20.5
HR	22.0
finance	23.0

Question 3

Show departments with more than 1 student

Student-id	name	age	department
1	Alice	20	IT
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	finance
5	Eve	22	HR

Query: `SELECT department,
COUNT(*) AS student_count
FROM students
GROUP BY department
HAVING COUNT(*) > 1;`

department	student_count
IT	2
HR	2

Question 4

Get all students whose age is between 21 and 23

Query: `SELECT student-id, name, age, department
FROM students
WHERE age BETWEEN 21 and 23;`

Question 5

List all the students in the IT or HR department who are older than 21

```
Query: SELECT student-id,  
             name,  
             age,  
             department  
        FROM Students  
       WHERE department IN ('IT', 'HR')  
             AND age > 21;
```

Student-id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

Question 6

Show total credits per departments, only for departments with more than 5 total credits

```
Query: SELECT department,  
             SUM(credits) AS total_credits  
        FROM Courses  
       GROUP BY department  
      HAVING SUM(credits) > 5;
```

department	Total_credits
IT	11

Question 7

List all Courses that do not have 4 credits

```
Query: SELECT course_id,  
              course_name,  
              department,  
              credits
```

```
FROM courses
```

```
WHERE credits <> 4;
```

Course_id	Course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

Question 8

Show the top 3 courses by credits in descending order

```
Query: SELECT course_id,  
              course_name,  
              credits  
FROM courses  
ORDER BY credits DESC  
LIMIT 3;
```


Course-id	Course-name	Credits
102	Python	4
103	Data Science	4
101	SQL Basics	4

Question 9

Get the maximum, minimum and average grade across all enrollments

Query: `SELECT max(grade) AS max_grade,
min(grade) AS min_grade,
AVG(grade) AS Avg_grade
FROM enrollments;`

max_grade	min_grade	avg_grade
90	78	84.6

Question 10

Count how many enrollments exist per course.

Query: `SELECT course-id,
COUNT(*) AS enrollment_count
FROM enrollments
GROUP BY course-id;`

Course - ID	enrollment - count
101	1
102	1
103	1
104	1
105	1

Question 11

Find the total salary and total bonus per department

Query : department,

sum(salary) AS total_salary,

sum(bonus) AS total_bonus

from salaries

Group by department;

department	total_salary	total_bonus
IT	122 000	10 500
HR	109 000	7500
finance	70 000	6000

Question 12

Show department when the average salary is above 55 000

Query: SELECT department,
AVG(salary) AS avg-salary
FROM Salaries
GROUP BY department
HAVING AVG(salary) > 55 000;

department	avg salary
IT	61 000
Finance	70 000

Question 13

List the employees whose salary plus bonus is greater than 60 000

Query: SELECT employee-id,
name,
salary,
bonus
FROM Salaries
WHERE salary + bonus > 60 000;

employee-id	name	salary	bonus	total-salary
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

Question 15

Show total and average budget per department, only include departments with average budget above 70 000.

Query: SELECT department,
SUM(budget) AS total-budget
AVG(budget) AS avg-budget
FROM projects
GROUP BY department
HAVING AVG(budget) > 70 000;

department	total-budget	avg-budget
IT	27 000	135 000
finance	80 000	80 000

Question 15

List all projects with the budgets between 50 000 and 120 000, excluding the marketing department.

Query: SELECT project-id,
project-name,
department,
budget
FROM projects
WHERE budget BETWEEN 50 000 AND
120 000
AND department <> 'marketing';

Project ID	Project Name	Department	Budget
1	AI App	IT	120000
2	Payroll System	Finance	80000
3	HR Portal	HR	50000